NOTE: The document identifier and heading has been changed on this page to reflect that this is a performance specification. There are no other changes to this document. The document identifier on subsequent pages has not been changed, but will be changed the next time this document is revised.

INCH-POUND

MIL-PRF-20/35H
21 April 1993
SUPERSEDING
MIL-C-20/35G
6 May 1988

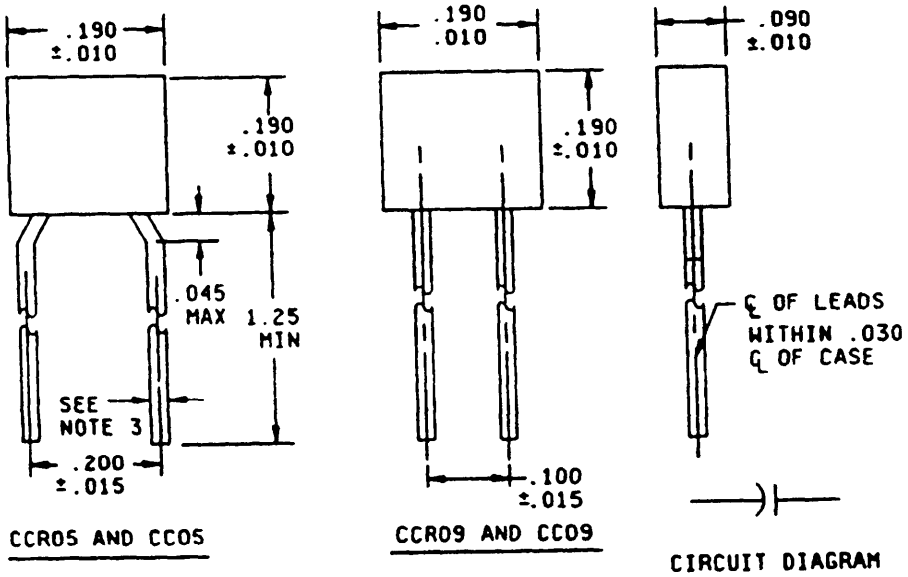
PERFORMANCE SPECIFICATION SHEET

CAPACITORS, FIXED, CERAMIC DIELECTRIC (TEMPERATURE COMPENSATING), ESTABLISHED AND NON-ESTABLISHED RELIABILITY, STYLES CCR05, CC05, CCR09, AND CC09.

Styles CC05 and CC09 are inactive for design after 19 September 1983. Use CCR05 and CCR09.

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-C-20.



CONFIGURATION WITHOUT STANDOFF

Inches	mm
.001	0.03
.004	0.10
.010	0.25
.015	0.38
.025	0.64
.028	0.71
.030	0.75
.045	1.15
.090	2.29
.100	2.54
.190	4.83
.200	5.08
1.250	31.75

FIGURE 1. Styles CCR05, CC05, CCR09, and CC09 capacitors.

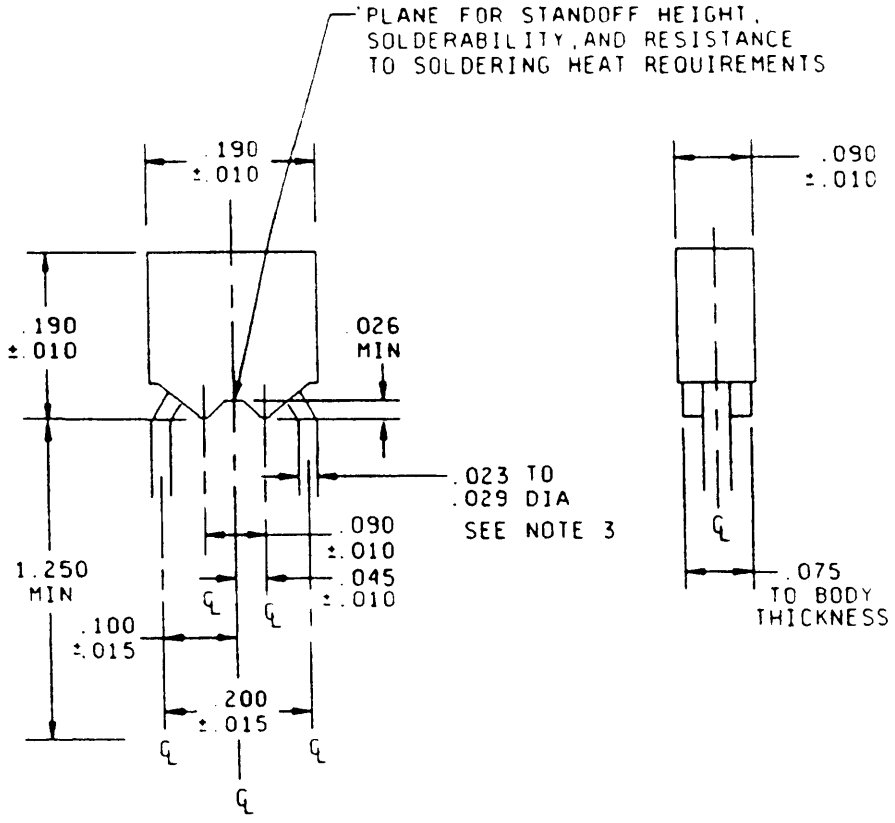
(M) denotes changes

FSC 5910

AMSC N/A

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OPTIONAL CONFIGURATION WITH STANDOFF (CCR05 ONLY)

NOTES:

1. Dimensions are in inches.
2. Metric equivalents are given for information only.
3. Lead diameter shall be .023 (0.58 mm) to .029 (0.74 mm).
4. Optional standoffs for style CCR05 shall be of triangular shape so that the standoffs will provide line contact with surface upon which the capacitor is mounted. The shape of the triangle is optional.
5. Thickness of standoffs shall be .075 (1.90 mm) to the body thickness.
6. Lead length may be a minimum of .625 (15.88 mm) for use in tape and reel automatic insertion equipment, when specified.
7. At the option of the user, the standoff configuration may be furnished as a replacement for the nonstandoff configuration of the same style.

FIGURE 1. Styles CCR05, CC05, CCR09 and CC09 capacitors - Continued.

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REQUIREMENTS:

Design and construction:

Dimensions and configuration: See figure 1.

Lead type: Radial.

Case type: Molded.

DC rated voltage: See table I.

Operating temperature range: -55°C to +125°C.

Characteristic: See table I.

Failure rate level (CCR05 and CCR09 only): M (1.0 percent), P (0.1 percent), R (.01 percent) or S (.001 percent).

Thermal shock and voltage conditioning (CCR05 and CCR09 only): In accordance with MIL-C-20.

Capacitance: Within tolerance specified (see table I).

Dissipation factor: In accordance with MIL-C-20.

Dielectric withstanding voltage: In accordance with MIL-C-20.

Body insulation: Test II.

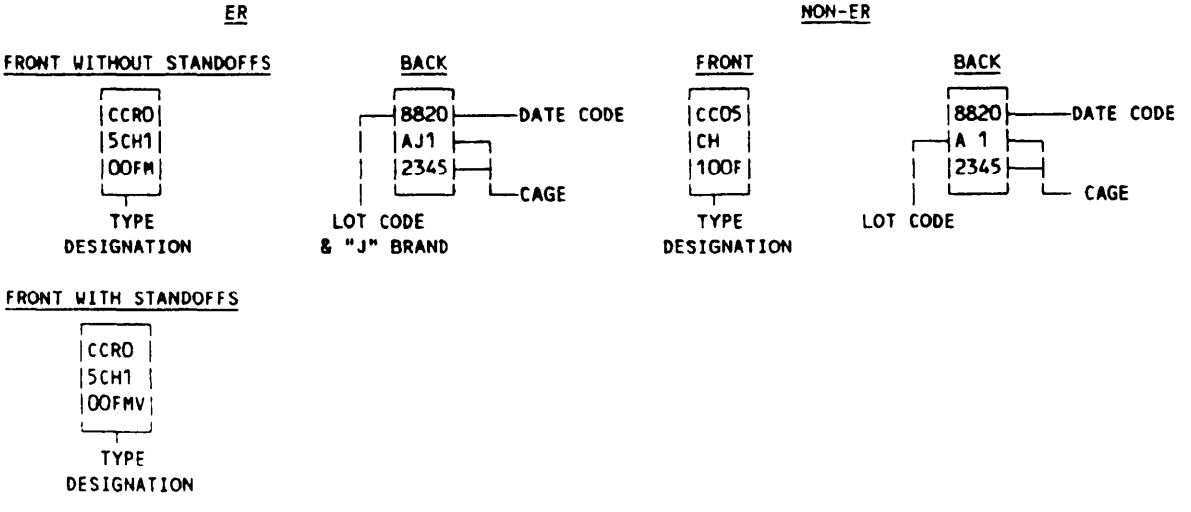
(H) Solderability: The leads shall be solderable up to .020 inch (0.51 mm) from the body egress.

Resistance to soldering heat: In accordance with method 210 of MIL-STD-202, test condition D.

Workmanship: For style CCR05 with standoff, exposed copper or bare leads shall be permitted to a maximum of .020 inch (0.51 mm) from the body egress. For style CCR05 without standoff, workmanship shall be in accordance with MIL-C-20.

Life: In accordance with MIL-C-20, operating condition 2.

Marking: Method I of MIL-STD-1285. At the option of the manufacturer, the marking may be placed on one side of the capacitor, or the marking may be placed on two lines as long as it is in the same order as shown in the following examples:



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TABLE I. Capacitor characteristics.

Type designation 1/ 2/	Rated voltage	Nom. cap.	Cap. tol.	Type designation 1/ 2/	Rated voltage	Nom. cap.	Cap tol.
	volts, dc	pF			volts, dc	pF	
3/							
CC-0-CX1R0---	200	1.0	BC	CC-0-CH180---	200	18	FGJ
CC-0-CX1R1---	"	1.1	"	CC-0-CG200---	"	20	"
CC-0-CX1R2---	"	1.2	"	CC-0-CG220---	"	22	"
CC-0-CX1R3---	"	1.3	"	CC-0-CG240---	"	24	"
CC-0-CX1R5---	"	1.5	"	CC-0-CG270---	"	27	"
CC-0-CX1R6---	"	1.6	"	CC-0-CG300---	"	30	"
CC-0-CX1R8---	"	1.8	"	CC-0-CG330---	"	33	"
CC-0-CX2R0---	"	2.0	"	CC-0-CG360---	"	36	"
CC-0-CK2R2---	"	2.2	"	CC-0-CG390---	"	39	"
CC-0-CK2R4---	"	2.4	"	CC-0-CG430---	"	43	"
CC-0-CK2R7---	"	2.7	BCD	CC-0-CG470---	"	47	"
CC-0-CK3R0---	"	3.0	"	CC-0-CG510---	"	51	"
CC-0-CK3R3---	"	3.3	"	CC-0-CG560---	"	56	"
CC-0-CK3R6---	"	3.6	"	CC-0-CG620---	"	62	"
CC-0-CK3R9---	"	3.9	"	CC-0-CG680---	"	68	"
CC-0-CJ4R3---	"	4.3	"	CC-0-CG750---	"	75	"
CC-0-CJ4R7---	"	4.7	"	CC-0-CG820---	"	82	"
CC-0-CJ5R1---	"	5.1	"	CC-0-CG910---	"	91	"
CC-0-CJ5R6---	"	5.6	"	CC-0-CG101---	"	100	"
CC-0-CJ6R2---	"	6.2	"	CC-0-CG111---	"	110	"
CC-0-CJ6R8---	"	6.8	"	CC-0-CG121---	"	120	"
CC-0-CJ7R5---	"	7.5	"	CC-0-CG131---	"	130	"
CC-0-CH8R2---	"	8.2	"	CC-0-CG151---	"	150	"
CC-0-CH9R1---	"	9.1	"	CC-0-CG161---	"	160	"
CC-0-CH100---4/	"	10	FGJ	CC-0-CG181---	"	180	"
CC-0-CH110---	"	11	"	CC-0-CG201---	"	200	"
CC-0-CH120---	"	12	"	CC-0-CG221---	"	220	"
CC-0-CH130---	"	13	"	CC-0-CG241---	"	240	"
CC-0-CH150---	"	15	"	CC-0-CG271---	"	270	"
CC-0-CH160---	"	16	"	CC-0-CG301---	"	300	"
CC-0-CG331---	"	330	"	CC-0-CG112---	100	1100	"
CC-0-CG361---	100	360	"	CC-0-CG122---	"	1200	"
CC-0-CG391---	"	390	"	CC-0-CG132---	"	1300	"
CC-0-CG431---	"	430	"	CC-0-CG152---	"	1500	"
CC-0-CG471---	"	470	"	CC-0-CG162---	"	1600	"
CC-0-CG511---	"	510	"	CC-0-CG182---	"	1800	"
CC-0-CG561---	"	560	"	CC-0-CG202---	50	2000	"
CC-0-CG621---	"	620	"	CC-0-CG222---	"	2200	"
CC-0-CG681---	"	680	"	CC-0-CG242---	"	2400	"
CC-0-CG751---	"	750	"	CC-0-CG272---	"	2700	"
CC-0-CG821---	"	820	"	CC-0-CG302---	"	3000	"
CC-0-CG911---	"	910	"	CC-0-CG332---	"	3300	"
CC-0-CG102---	"	1000	"				

See footnotes on next page.

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TABLE I. Capacitor characteristics - Continued.

1/ Complete type designation will include the following:

1st dash: Symbol "R" (for styles CCR05 and CCR09) or dash will be deleted (for styles CC05 and CC09).
2nd dash: Applicable style number (5 for .200 ±.015 lead spacing and 9 for .100 ±0.015 lead spacing).
3rd dash: Applicable capacitance tolerance symbol.
4th dash: Applicable failure rate level symbol (CCR05 and CCR09 only) or dash will be deleted (for styles CC05 and CC09).
5th dash: Symbol "V" (for standoff mounting configuration) or dash will be deleted for no standoff.

2/ With respect to characteristics in previous issues of this specification sheet: For capacitance values of 2 pF or less, characteristic CX is interchangeable with characteristic CH; for capacitance values of 2.2 pF through 3.9 pF, characteristic CK is interchangeable with characteristic CH; for capacitance values of 4.3 pF through 7.5 pF, characteristic CJ is interchangeable with characteristic CH; and for capacitance values of 11 pF through 18 pF, characteristic CH is interchangeable with characteristic CG. These parts shall be stocked under the same applicable NSN's.

3/ Characteristic CX: Not practically measurable.

4/ Capacitance tolerance: (F) ±1 percent and (G) ±2 percent.

Temperature coefficient and capacitance drift: In accordance with MIL-C-20. For capacitors with values of 18 pF or less, the temperature-coefficient tolerance shall be determined in accordance with table II.

TABLE II. Temperature coefficient tolerances.

Permissible capacitance change from capacitance at +25°C in ppm/°C					
Temperature	Characteristic				
	CX	CK	CJ	CH	CG
+125°C	1/	± 250 ppm/°C	± 120 ppm/°C	± 60 ppm/°C	± 30 ppm/°C
-55°C 2/	1/	+246.25 -326.25	+116.25 -166.25	+55.00 -91.25	+27.50 -53.75

1/ Not practically measurable.

2/ The ppm/°C values for -55°C were calculated by dividing ppm by negative 80°C.

CONCLUDING MATERIAL

Custodians:
Army - ER
Navy - EC
Air Force - 85

Review activities:
Navy - OS
Air Force - 99
DLA - ES

User activities:
Navy - AS, MC, SH

Preparing activity:
Air Force - 85

Agent:
DLA - ES

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